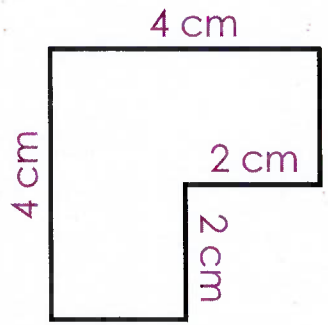
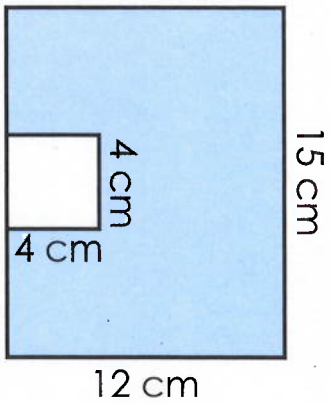
GM5aXv1

Directions: Calculate the area and perimeter. Show your work.



area: _____ perimeter: _____

Directions: Find the area of the shaded region.



area: _____

GM5aX7.8

Level 5a

bestbrains LEARNING CENTERS

Name Ojas Raj

Date / /

Time ____ : ____ TO ____ : ____

Week 24

X1

Fractions: All Operations **Statistical Understanding**

A florist created 3 different bouquets.

Bouquet	Number of Flowers	Roses	Tulips	Lilies
1	36	$18\frac{3}{6}$	$6\frac{6}{36}$	$12\frac{1}{3}$
2	42	$15\frac{5}{14}$	$21\frac{1}{2}$	$6\frac{2}{14}$
3	50	$10\frac{1}{5}$	$15\frac{3}{10}$	$25\frac{2}{4}$

R: 18 + 15 = 10 = 43

T: 16 21 15 = 42

L: 25 + 12 = 6 = 43

How many of each type of flower is used in all the bouquets?

Roses 43 Lilies 43

Tulips 42

GM5aX1.1

Directions: Find the LCD of each set of numbers.

1. $\frac{1}{3}$:
   $\frac{4}{5}$:
   $\frac{5}{6}$:

   The LCD of 3, 5 and 6 is $\boxed{30}$

2. $\frac{1}{2}$:
   $\frac{2}{3}$:
   $\frac{5}{9}$:

   The LCD of 2, 3 and 9 is $\boxed{18}$

Directions: Find the average of $3\frac{2}{3}$, $1\frac{5}{24}$ and $7\frac{5}{8}$.

$3+1+7=11$

$\frac{\cancel{2}16}{\cancel{3}24}+\frac{5}{24}+\frac{\cancel{5}15}{\cancel{8}24}=\frac{36}{24}=1\frac{12}{24}=1\frac{1}{2}$

$11+1\frac{1}{2}=12\frac{1}{2}$

$12\frac{1}{2}\div\frac{3}{1}=$

$12\frac{1}{2}\times\frac{1}{3}=$

$\frac{25}{2}\times\frac{1}{3}=\frac{25}{6}=4\frac{1}{6}$

$4\frac{1}{6}$

Directions: Which statement is true?

a. The LCM of 12 and 16 is 4.

b. The LCM of 45 and 30 is 90.

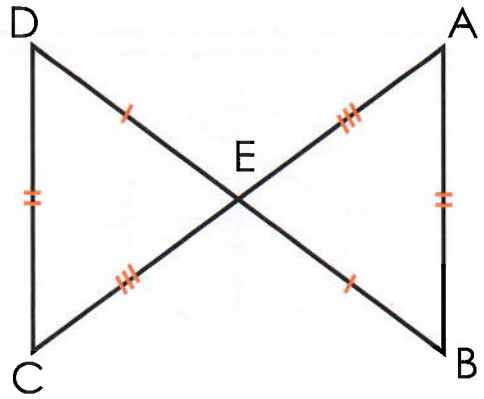
Directions: Use factor trees to find the LCM.

45

60

LCM = _____

Which statement is true?



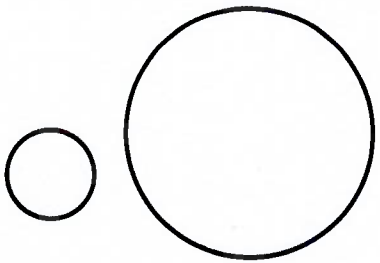
a. <AEB ≅ <DCE

b. <D ≅ <C

c. $\overline{AB} \cong \overline{CE}$

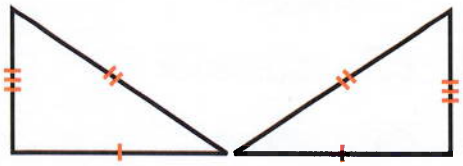
d. $\overline{AE} \cong \overline{CE}$

Directions: State whether the shapes are congruent or not.



☐ congruent

☐ not congruent



☐ congruent

☐ not congruent

GM5aX7.6

Directions: Determine the measure of each angle.

The sum of two angles is $6\frac{3}{5}°$. Both angles are identical.

$6\frac{3}{5} \div 2 =$

$\frac{33}{5} \times \frac{1}{2} = \frac{33}{10} = 3\frac{3}{10}$

$3\frac{3}{10}$

What is $\frac{1}{6}$ of 21?

$\frac{1}{6} \times \frac{21}{1} = \frac{21}{6} = 3\frac{3}{6} = 3\frac{1}{2}$

a. $3\frac{1}{3}$

b. $4\frac{2}{3}$

c. $5\frac{1}{3}$

d. $3\frac{1}{2}$

GM5aX1.3

At a carnival, $\frac{1}{8}$ of the people are men and $\frac{1}{4}$ of the people are woman. The remaining people are children. If there are 136 people total, how many are children?

$136 \times \frac{1}{8} = \frac{136}{8} = 17$

$17 \times 2 = 34$

$17 + 34 = 51$

$136 - 51 = 85$

a. 34

b. 17

c. 51

d. 85

Lamar is $5\frac{3}{4}$ feet tall. Jackson is $1\frac{1}{3}$ feet taller than Lamar. How tall are the boys combined?

$5\frac{3}{4} + 1\frac{1}{3} =$

$\frac{69}{12} + \frac{16}{12} = \frac{85}{12} = 7\frac{1}{12}$ ft.

a. $4\frac{5}{12}$ feet

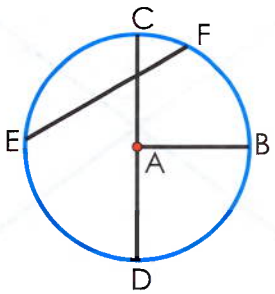
b. $7\frac{1}{12}$ feet

c. $12\frac{5}{6}$ feet

d. $6\frac{3}{4}$ feet

GM5aX1.4

Directions: Identify the diameter in the diagram below.



a. $\overline{EF}$

b. $\overline{AB}$

c. $\overline{AD}$

d. $\overline{CD}$

Draw a circle and label each part correctly.

$\overline{FA}$ = radius

F = center

$\overline{BD}$ = diameter

$\overline{CE}$ = chord

GM5aX7.5

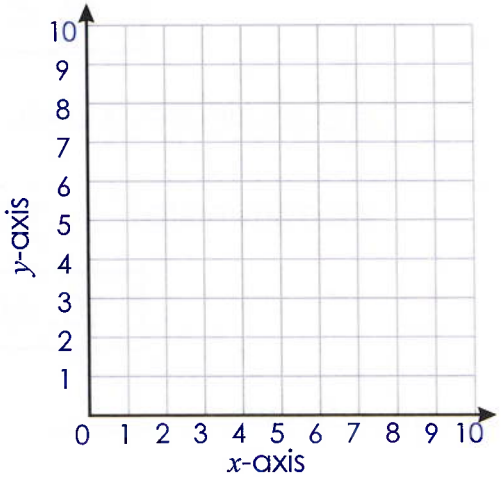
Directions: Graph the points. Then determine the shape and circle it.

A. (3, 5)

B. (7, 6)

C. (6, 2)

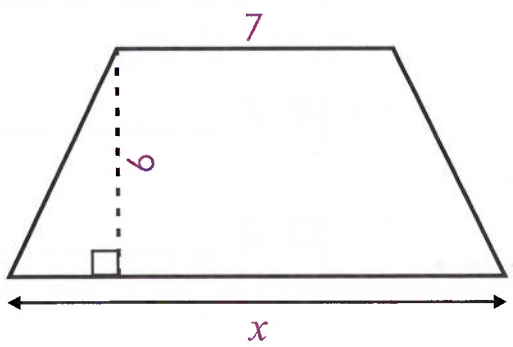
D. (2, 1)



parallelogram rectangle rhombus square

Directions: Find the value of x if the area of the trapezoid is 54 inches^2.

$$\text{Area} = \frac{(\text{base 1} + \text{base 2})}{2} \times \text{height}$$



Statistical Understanding

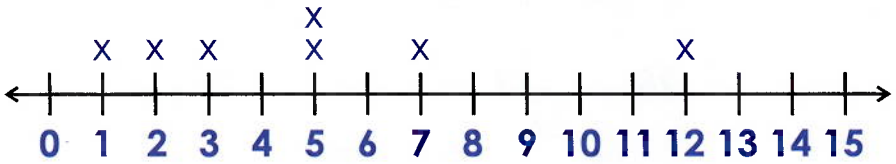
Any set of statistical data can be described by its center and spread. The measures of the **center** of the data are the mean, median and the mode. The **spread** is referred to as the range.

Mode = most frequently occurring value
Mean = add all values and divide by total number
Median = the middle value when numbers are in order from least to greatest
Range = the difference between the largest and smallest values

There are multiple measures of spread, the most common being range. Let's use the frequency table and plotted information to find all the data.

At Cambridge High school, the hockey team is undefeated. Interpret the graph.

Team Member	Mark	Rob	Saol	Ruben	Arun	Ahmad	Jo
Score	3	2	1	5	5	12	7

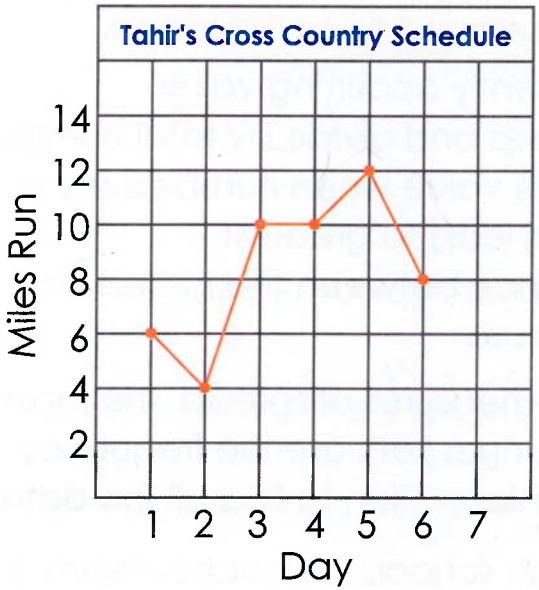


Calculate the mean, median, mode and range.

Mean: 5 **Mode:** 5

Median: 5 **Range:** 11

Directions: Find the mean, median and mode of the data represented.



Mean = $8\frac{2}{6} = 8\frac{1}{3}$

Median = 9

Mode = 10

In the Chukaroo Forest, there is a group of sequoia trees. The diameters of the tree trunks in feet are listed.

24	34	21	
26	17	33	29
28	26	27	32

Mean = 27 Median = 26 Mode = 26

GM5aX1.6

Directions: Describe the move on each axis.

(3, 5)

x-coordinate: 3 units to the _____

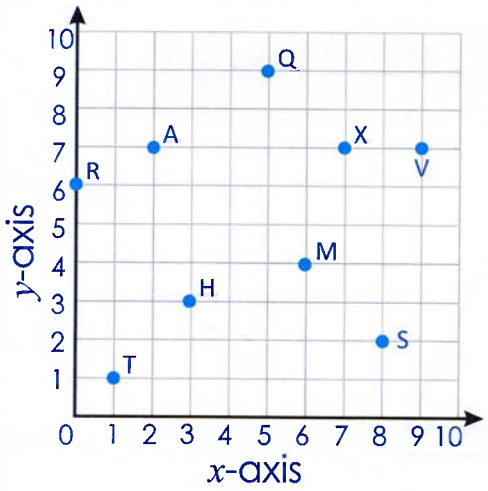
y-coordinate: ____ units _____

(0, 4)

x-coordinate: _____

y-coordinate: _____

Directions: Determine what letter is located at each ordered pair.



(1) (6, 4) _____

(2) (8, 2) _____

(3) (0, 6) _____

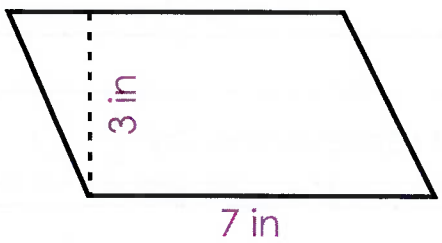
GM5aX7.3

Directions: Solve and show your work.

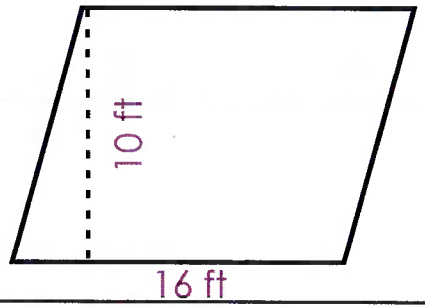
Draw a triangle with a base of 3 cm and a height of 7 cm. Calculate the area.

Directions: Calculate the area.

1



2



Directions: Find all the measures of the center and spread.

Vikings

Season	Wins
1	4
2	9
3	13
4	9
5	8
6	9
7	4

Bulldozers

Season	Wins
1	6
2	8
3	3
4	3
5	3
6	10
7	9

(a) Mean = 8 Mode = 9

Median = 9 Range = 9

Mean = 6 Mode = 3

Median = 6 Range = 7

(b) Which team had the better record over the past 7 seasons? Justify your answer.

The Vikings had more wins.

Directions: Select the statistical questions.

a. How old are you?

b. What percent of teachers work during the summer months?

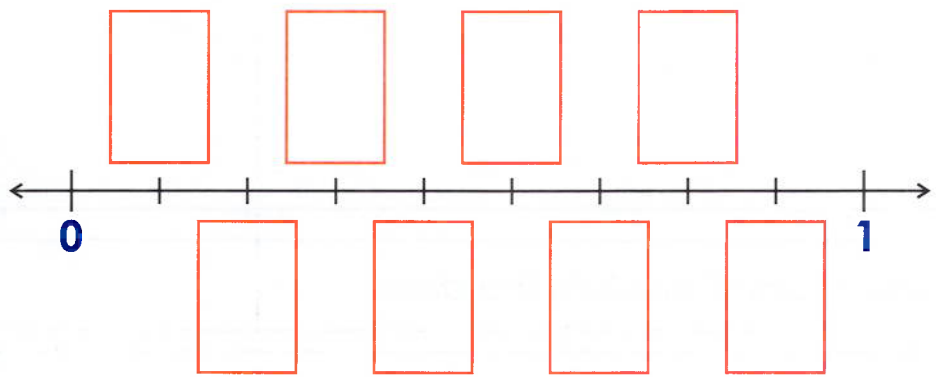
c. What is the height of your teacher?

d. What is the favorite color of each student in your class?

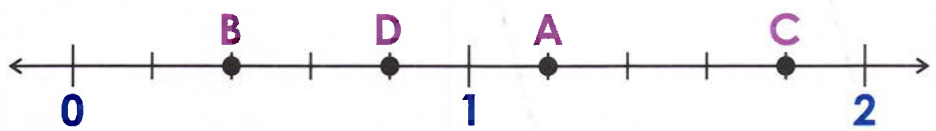
e. What is the average height of a building in New York City?

Level Review

Directions: Label the number line.



Directions: Circle the point represented by $\frac{4}{5}$.



Level 5a

bestbrains LEARNING CENTERS

Name

Date / /

Time ____ : ____ TO ____ : ____

Week 24

X2

Directions: Determine which set of data results in the following.

mean: 14 **mode: 12**

median: 12 **range: 7**

a. 12, 17, 19, 12, 13, 16, 12

b. 15, 13, 19, 18, 20, 18, 16

c. 10, 25, 16, 18, 32, 14, 10

d. 16, 12, 18, 12, 11, 17, 12

GM5aX6.8

Fractions: All Operations **Statistical Understanding**

In Mrs. Smith's class, $\frac{1}{3}$ of the students have a brother. Of these students, $\frac{3}{4}$ also have a sister. What fraction of the students have both a brother and a sister?

What fraction of students with a brother do not have a sister?

GM5aX2.1

Which improper fraction is equal to $4\frac{3}{8}$?

$4\frac{+3}{\times 8} = \frac{35}{8}$

a. $\frac{32}{8}$

b. $\frac{20}{8}$

c. $\frac{35}{8}$

d. $\frac{12}{8}$

Directions: Solve.

$5 \div \frac{4}{9} =$

$5 \times \frac{9}{4} = \frac{45}{4} = 11\frac{1}{4}$

$11\frac{1}{4}$

$9\frac{\cancel{4}\,32}{\cancel{5}\,40} - \frac{\cancel{2}\,10}{\cancel{8}\,40} =$

$\frac{392}{40} - \frac{10}{40} = \frac{382}{40} = 9\frac{22}{40}$

$= 9\frac{11}{20}$

$9\frac{11}{20}$

GM5aX2.2

Directions: Determine if the mean would be skewed.

Color	Frequency
Pink	𝍸 𝍸 ||
Blue	𝍸 𝍸 𝍸
Red	𝍸 𝍸 |||
Green	𝍸 𝍸
White	𝍸 ||

☐ skewed

☐ not skewed

Explain.

Directions: Determine if the data is statistical or not.

Justify your answer.

Range = 0

Median = 3

Mode = 3

Mean = 3

GM5aX6.7

Directions: Answer each question.

Movie	Time(min)
Fairy Fairy	124
Dark Dreams	96
Super Nova	100
Gone	96

1. What is the mode?

2. What is the average length of the movies?

Directions: Determine the outlier in the data below.

5 3 6 10 1 27 8 9 7 3 3

Lengths of Test Worms (in)

How would the outlier effect the center and spread?

GM5aX6.6

Directions: Use the butterfly method to solve.

1. $\frac{7}{15} - \frac{3}{8} = \frac{11}{120}$

56 45 120 120

$\frac{11}{120}$

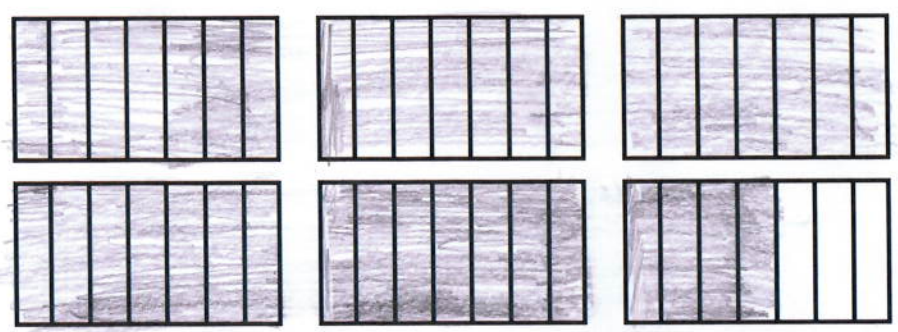
2. $\frac{3}{4} + \frac{1}{8} = \frac{7}{8}$

6 8

$\frac{7}{8}$

Directions: Use the model to determine the missing value.

$5\frac{4}{7} + \boxed{\frac{3}{7}} = 6$



GM5aX2.3

Directions: Which solution has the greatest value?

a. $\frac{6}{10}^{18} + \frac{3}{5}^{18} + \frac{2}{3}^{20} = \frac{56}{30} = 1\frac{26}{30} = 1\frac{13}{15}$

b. $9\frac{1}{4}^{5} - 3\frac{4}{5}^{16} = \frac{185}{20} - \frac{76}{20} = \frac{109}{20} = 5\frac{9}{20}$

c. $3\frac{3}{5} \times 2\frac{1}{3} = \frac{18}{15} \times \frac{7}{3} = \frac{126}{45} = 2\frac{36}{45} = 2\frac{12}{15}$

d. $\frac{6}{10} \div 3 = \frac{6}{10} \times \frac{1}{3} = \frac{6}{30} = \frac{1}{5}$

Directions: Find the mean, median, mode and range for each set of numbers.

3	18	12	7	6	12	5

Mean:

Median:

Mode:

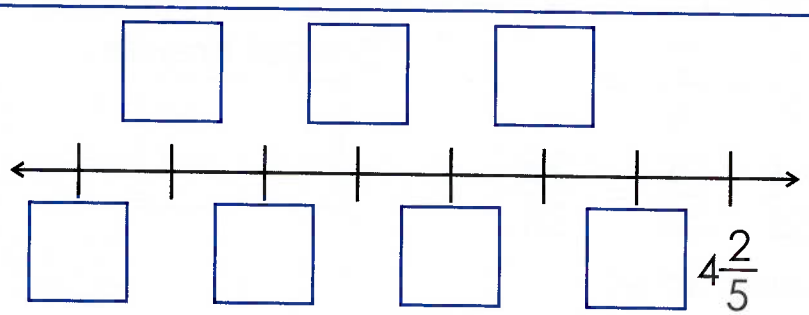
Range:

Directions: Determine if the question is statistical or not.

What is the height of the oak tree behind Caleb's house?

☐ statistical ☐ non statistical

Directions: Divide the number line into 7 equal parts. Label each portion of the line.



$4\frac{2}{5} \div 8 =$ ☐

Directions: Solve.

$3\frac{1}{4} \times \frac{5}{6} =$	$8\frac{9}{10} - 1\frac{3}{8} =$

GM5aX6.4

Statistical Understanding

When using the mean to find your center, an outlier can dramatically skew the results. An outlier is a piece of data that lies outside most other values.

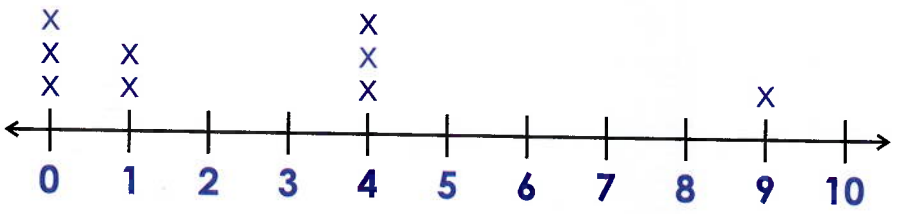
Ex. Find the average hourly rate for the employees at Row Boat House.

$10, $15, $12, $15, $50

Mode: $12
Median: $15
Mean: $19.80

Here we can see the three measurements of center. Because of the $50 outlier, the mean has been skewed. Therefore, the best calculation of the center would be the median and/or mode.

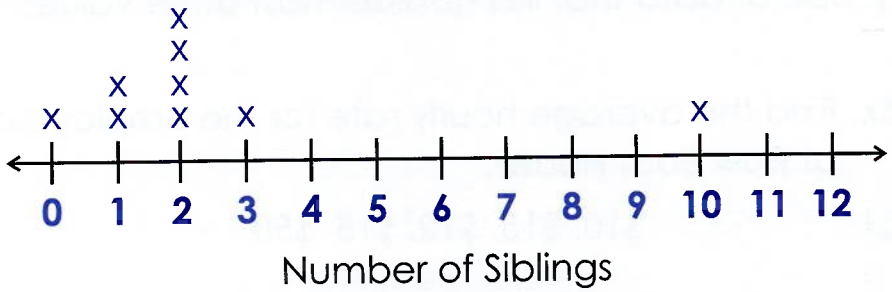
By looking at this data, determine if it is skewed.



☐ skewed ☐ not skewed

GM5aX2.5

Directions: Look at the graph to determine if the mean would be skewed.



☐ skewed ☐ not skewed

Directions: Determine the average test score.

The sixth grade students in Mr. Broker's class took a math test and received the following scores.

96, 89, 80, 94, 83, 91, 95, 86, 92, 84

96 + 89 + 80 + 94 + 83 + 91 = 533

95 + 86 + 92 + 84 = 357

533 + 357 = 890

890 ÷ 10 = 890 × 1/10 = 890/10 = 89

Was there an outlier?

☐ Yes ☐ No

GM5aX2.6

Directions: Describe the mistake in the solved problem.

$$\frac{6}{9} + \frac{2}{4} =$$

$$\frac{24}{36} + \frac{8}{36} = \frac{32}{36}$$

Correct Answer

Explain error:

Directions: Determine the value of the equation.

$$\left(3\frac{1}{2} \div 2\right) \div \frac{4}{7}$$

a. $3\frac{1}{16}$

b. $3\frac{1}{4}$

c. $3\frac{3}{8}$

d. $3\frac{2}{16}$

GM5aX6.3

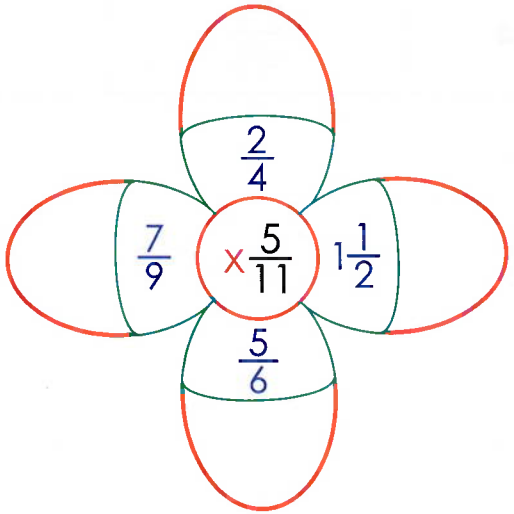
Directions: Solve and circle the answer that does not belong.

a. $\frac{5}{7} \times 2 =$

b. $\frac{2}{3} - \frac{1}{10} =$

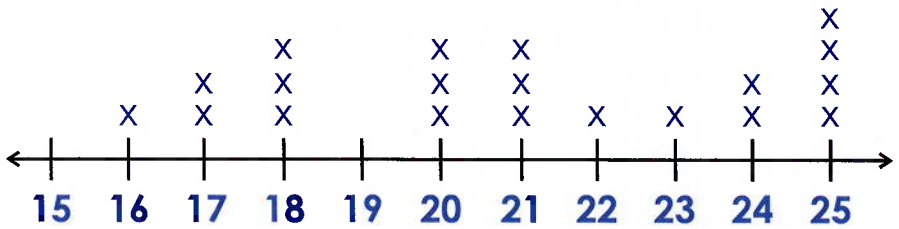
c. $4\frac{16}{21} - 3\frac{1}{3} =$

Directions: Multiply the number in the middle by each petal.



GM5aX6.2

Directions: Determine the measures of center.



Number of Minutes Spent Studying

20	20	25
Mean	Median	Mode

Directions: Look at the graphs to determine if the mean would be skewed.

Szymon surveyed his friends to find out their favorite pizza topping. Below are the results.

P, V, M, C, C, V, C, M, P, P, P, V, C, M

Topping	Frequency
Mushroom	3
Pepperoni	4
Cheese	4
Vegetable	3

☐ skewed

☒ not skewed

GM5aX2.7

Directions: If the median of a set of numbers is 17, determine the invalid value of x.

16, 17, x, 20, 18

a. $x = 10$

b. $x = 13$

c. $x = 21$

d. $x = 17$

Level 5a

bestbrains LEARNING CENTERS

Name

Date / /

Time ____ : ____ TO ____ : ____

Week 24

X6

Fractions: All Operations **Statistical Understanding**

Directions: The table below represents equivalent fractions. Identify the row that contains the error.

	Fraction A	Fraction B
Row 1	$\frac{7}{56}$	$\frac{1}{8}$
Row 2	$\frac{14}{28}$	$\frac{1}{2}$
Row 3	$\frac{13}{21}$	$\frac{3}{7}$
Row 4	$\frac{15}{45}$	$\frac{1}{3}$

a. Row 1

b. Row 2

c. Row 3

d. Row 4

Level 5a

bestbrains LEARNING CENTERS

Name

Date / /

Time ____ : ____ TO ____ : ____

Week 24

X3

Directions: Consider the following sample set: 5, 13, 6, 9, 5, 5, 21, 15, 13, 5, 11, 4, 2, 16, 13, 7. Choose all the true statements.

a. Mean is 10

b. Range is 19

c. Mean is 9

d. Mode is 5

GM5aX5.8

Fractions: All Operations | **Statistical Understanding**

A recipe calls for $2\frac{1}{2}$ cups of chocolate chips. You only have $1\frac{3}{8}$ cups of chocolate chips. How many cups of chocolate chips are you short?

a. $\frac{1}{8}$ of a cup

b. $1\frac{2}{8}$ cups

c. $1\frac{1}{8}$ cups

d. $\frac{2}{8}$ of a cup

GM5aX3.1

Directions: Represent each problem as repeated addition. Then solve.

1. $5 \times \frac{2}{8} =$ _____ = ☐

2. $\frac{5}{4} \times 3 =$ _____ = ☐

3. $4 \times \frac{3}{8} =$ _____ = ☐

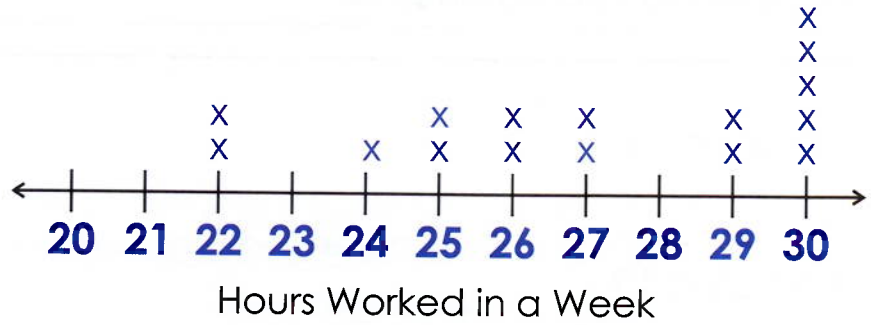
Directions: Evaluate when $x = \frac{3}{7}$ and $y = \frac{6}{5}$.

1. $y \div x =$

2. $x + y =$

GM5aX3.2

Directions: Determine the measures of center.



Center: _____ _____ _____

mean median mode

Directions: Look at the graphs to determine if the mean would be skewed.

Ramona surveyed her friends to find out their favorite day of the week. Below are the results.

F, F, Sa, Sa, Su, F, Th, F, F, Sa

Day	Frequency
Thursday	
Friday	
Saturday	
Sunday	

☐ skewed

☐ not skewed

GM5aX5.7

At the petting zoo there are many llamas. The height of each llama in inches is listed.

60	47	66	
61	50	66	65
58	61	66	

Mean =

Median =

Mode =

Identify the statement that is non statistical and rewrite it to be statistical.

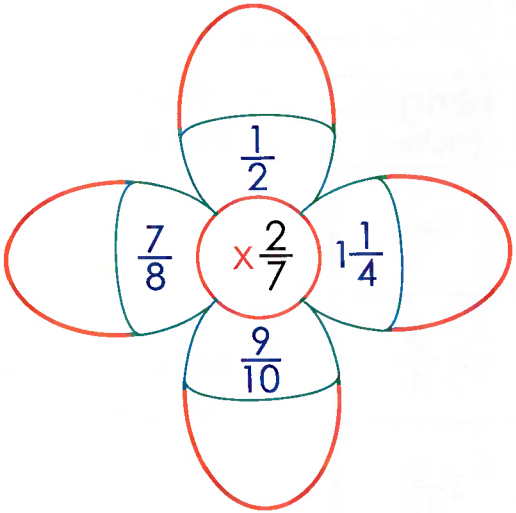
a. The tallest building is 2, 717 ft.

b. The average amount of candies in a bag is 25 pieces.

GM5aX5.6

Describe the steps you would use to solve the subtraction problem $3\frac{3}{4} - 1\frac{7}{8}$.

Directions: Multiply the number in the middle by each petal.



GM5aX3.3

A recipe for cookies requires $\frac{3}{4}$ of a cup of flour. If you are making half of a batch, how much flour do you need?

a. $1\frac{1}{2}$ cups

b. $\frac{1}{4}$ of a cup

c. $\frac{1}{2}$ of a cup

d. $\frac{3}{8}$ of a cup

Directions: The table shows the length of three different strings.

String	Length (inches)
String A	$1\frac{1}{2}$
String B	$8\frac{3}{4}$
String C	$1\frac{6}{10}$

1. What is the combined length of string A and B?

2. How much longer is string B than string C?

GM5aX3.4

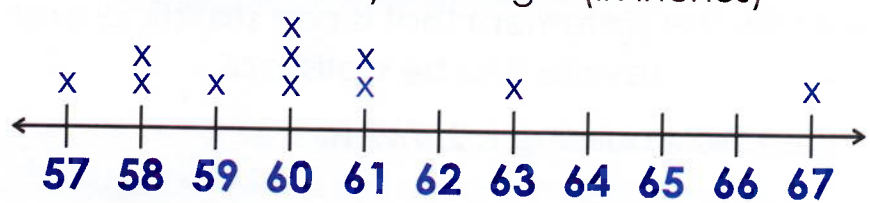
Statistical Understanding

The table represents the class average on a test for each of Mrs. Smith's classes.

Period	Class Average
1	88 %
2	91.2%
4	79%
5	81%

Is this data statistical? Explain.

Basketball Player's Height (in inches)



Based on the data above, what conclusions can we draw about basketball players and their height?

a. Most players are taller than the average

b. One person's height affects the rest of the data

c. Most players are under 60 inches

GM5aX5.5

On a necklace there are 25 pieces. $\frac{3}{5}$ of them are blue. How many pieces on the necklace are blue?

Siona plays soccer. She recorded her practice time in the chart below. How much longer did she practice on Monday compared to Tuesday?

Day	Mon.	Tues.	Wed.	Thurs.
Time (in hours)	$\frac{16}{24}$	$\frac{3}{9}$	$\frac{1}{18}$	$\frac{1}{11}$

a. $\frac{2}{3}$ of an hour

b. $\frac{1}{3}$ of an hour

c. $\frac{1}{9}$ of an hour

d. $\frac{2}{9}$ of an hour

GM5aX5.4

Statistical Understanding

Directions: Determine if the data is statistical or not.

The following are test scores from a high school science class.

10, 11, 11, 12, 13, 14, 15, 17, 23

Based on the information you determined:

Mean = 14 Median = 13

Mode = 11 Range = 13

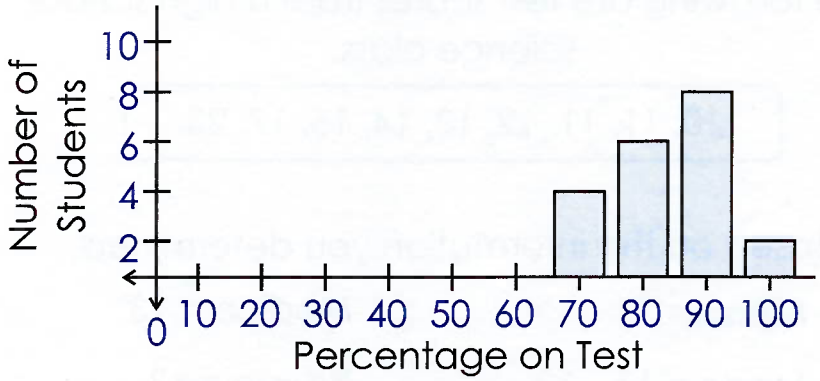
Is this information statistical? **Yes/No**

Directions: The numbers listed are the number of points scored in the last 6 basketball games. Describe the average amount of points scored in each game.

44, 38, 54, 48, 26, 36

GM5aX3.5

Directions: Determine all the measures of the center and spread.



a. How many students took the test?

b. What is the average score?

c. Could you determine the average just by looking at the graph? Explain.

GM5aX3.6

Directions: Convert each improper fraction to a mixed number.

1. $\frac{13}{5} =$

2. $\frac{29}{6} =$

3. $\frac{47}{6} =$

4. $\frac{59}{9} =$

Directions: Solve.

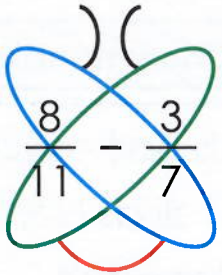
$7 \div \frac{3}{4} =$

$\frac{5}{8} \times \frac{2}{3} =$

GM5aX5.3

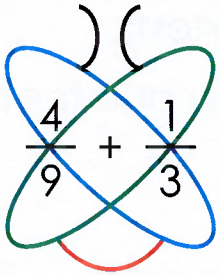
Directions: Use the butterfly method to solve.

1



$\frac{8}{11} - \frac{3}{7} =$

2

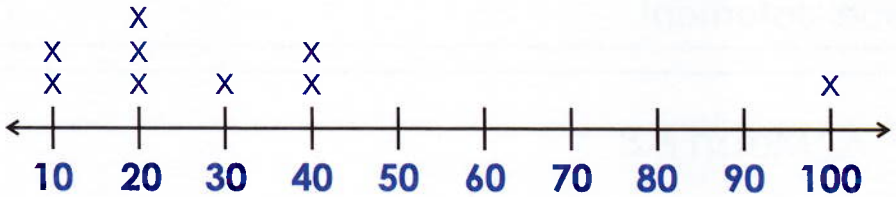


$\frac{4}{9} + \frac{1}{3} =$

Directions: Find the average of $1\frac{9}{11}$, $\frac{3}{5}$ and $\frac{12}{5}$.

GM5aX5.2

Directions: Determine if the mean is skewed.



Number of text messages sent in 24 hours

☐ skewed ☐ not skewed

What measure of center would be best to determine the average? Explain.

Directions: Find the mean, median, mode and range for each set of numbers.

39, 85, 25, 85, 37

Mean: **Mode:**

Median: **Range:**

GM5aX3.7

Level 5a

bestbrains LEARNING CENTERS

Name

Date / /

Time ____ : ____ TO ____ : ____

Week 24

X5

Directions: Consider the following sample set: 3, 12, 5, 18, 11, 8, 10, 6, 14, 11, 4, 11, 5, 12, 5, 13. Pick the true statement.

a. Mean is 8

b. Mode is 12

c. Median is 5

d. Range is 15

GM5aX3.8

Fractions: All Operations | **Statistical Understanding**

Peter is 5 feet 6 inches tall. Nora is half as tall as Peter. Cameille is $1\frac{1}{3}$ times as tall as Nora. Ronaldo is 62 inches tall. Order each person from tallest to shortest.

Remember there are 12 inches in one foot.

GM5aX5.1

Level 5a

bestbrains LEARNING CENTERS

Name

Date / /

Time ____ : ____ TO ____ : ____

Week 24

X4

Directions: Graph the data and find the mean, median and mode.

59, 60, 51, 60, 55, 50, 60, 50, 58, 57, 51, 60, 51, 58, 60

Mean:

Median:

Mode:

GM5aX4.8

Fractions: All Operations **Statistical Understanding**

Use your knowledge or fractions to determine which answer is more than 1 without solving.

a. $\frac{2}{4} \times \frac{3}{4}$

b. $\frac{2}{8} \times \frac{7}{9}$

c. $3\frac{1}{2} \div \frac{2}{3}$

d. $\frac{1}{3} + \frac{1}{2}$

Explain your answer.

GM5aX4.1

The total perimeter of a square shaped sandbox is $8\frac{1}{3}$ ft. What is the area of the sandbox?

The table shows the length of three different wingspans.

Bird	Wingspan
A	$8\frac{7}{8}$ in
B	$12\frac{1}{2}$ in
C	$3\frac{4}{9}$ in

a. What is the combined wingspan of all birds?

b. What is the difference between the longest and shortest wingspan?

GM5aX4.2

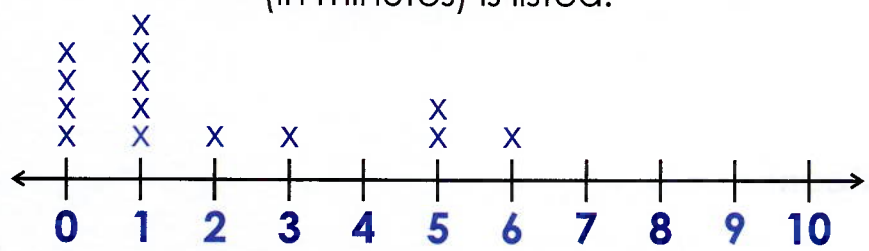
A survey of all freshmen students yielded a mean of 88 and a range of 62.

Is this information statistical?

Explain.

Directions: Look at the graph to determine if the mean would be skewed.

The average wait time at a call center (in minutes) is listed.



Which is true?

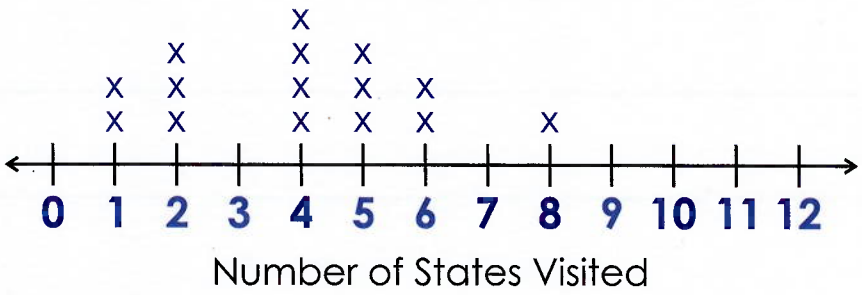
a. Most customers wait less than 2 minutes to speak to a representative.

b. The average wait time is 1 minute.

c. The data is skewed by an outlier.

GM5aX4.7

Directions: Look at the graph to determine if the mean would be skewed.



☐ skewed ☐ not skewed

Justify your answer.

_____

_____

Directions: Select all of the statistical questions.

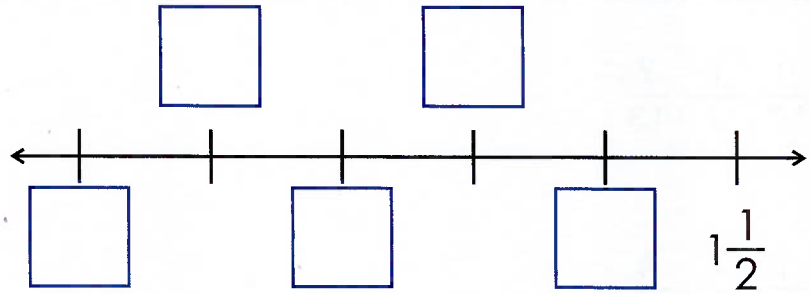
a. What percent of people like watermelon?

b. How long have you owned a dog?

c. How much money did you spend?

d. How old are the kids on this street?

Directions: Divide the number line into 6 equal parts. Label each portion of the line.



$1\frac{1}{2} \div 6 =$ ☐

Directions: Solve then determine if the answer is greater than or less than $\frac{1}{2}$.

1. $\frac{5}{6} - \frac{2}{5} =$ ☐ greater than $\frac{1}{2}$ ☐ less than $\frac{1}{2}$

2. $\frac{3}{8} - \frac{1}{9} =$ ☐ greater than $\frac{1}{2}$ ☐ less than $\frac{1}{2}$

Directions: In which arrangement are the fractions listed from least to greatest?

a. $\frac{8}{17}, \frac{1}{2}, \frac{7}{13}$

b. $\frac{1}{2}, \frac{7}{13}, \frac{8}{17}$

c. $\frac{8}{17}, \frac{7}{13}, \frac{1}{2}$

d. $\frac{1}{2}, \frac{8}{17}, \frac{7}{13}$

e. $\frac{7}{13}, \frac{1}{2}, \frac{8}{17}$

Directions: Explain the difference between statistical and non-statistical data.

Directions: Find the average number of text messages sent per day across a week. Hint: there are 7 days in a week.

	M	T	W	Th	F	Sa	Su
Am	6	3	8	9	10	15	20
Pm	11	18	15	20	20	25	30